

DETAILED PROJECT PLAN

Enterprise Learning Platform – Skill & Career Guidance System

1. Project Overview

The **Enterprise Learning Platform – Skill & Career Guidance System** is a multi-role digital learning and career development platform designed to connect students, mentors, administrators, courses, assessments, career roadmaps, internships, achievements, and certifications within a single system.

The platform provides separate experiences for:

Students

Students can:

- Discover and enroll in courses
- Track learning progress
- Complete lessons
- Take quizzes
- Submit assignments
- Monitor skill development
- Follow career roadmaps
- Receive recommendations
- Earn achievements
- Obtain certificates
- Access internship opportunities
- Interact with mentors

Mentors

Mentors can:

- Create and manage courses
- Manage lessons
- Monitor students

- Create assignments
- Review submissions
- Create and manage quizzes
- Grade student work
- Schedule learning sessions
- Monitor enrollments
- Communicate with students

Administrators

Administrators can:

- Manage users
 - Manage roles
 - Approve courses
 - Manage platform content
 - Monitor complaints
 - Manage internships
 - Review audit logs
 - Monitor system activity
 - Manage platform-level settings
 - Monitor overall platform performance
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2. Overall Project Goal

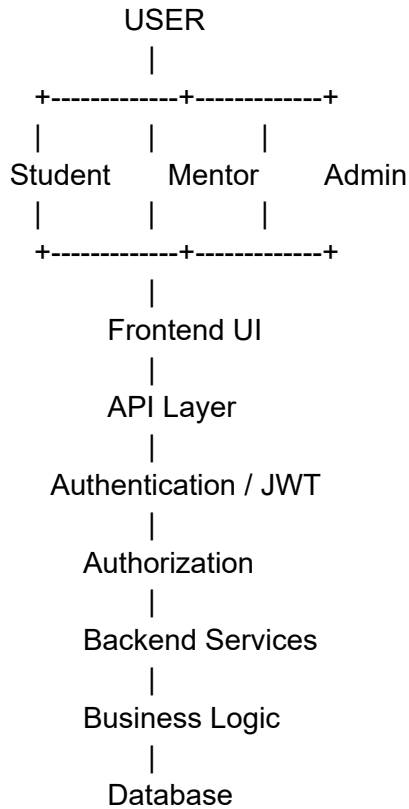
The primary goal is to develop a **secure, scalable, role-based learning and career guidance platform** that provides personalized learning experiences for students while giving mentors and administrators the tools required to operate and manage the platform.

The system should provide:

Secure Authentication + Personalized Learning + Course Management + Assessments + Career Guidance + Mentorship + Administration

3. Project Architecture

The overall system follows a layered architecture.



The architecture allows the platform to support multiple user roles while maintaining separation of permissions and data.

5. Milestone 1 – Platform Foundation & Authentication

Objective

Establish the initial platform structure and secure authentication system.

5.1 Landing Page

Develop the public landing page containing:

- Logo
- Navigation

- Home
- Features
- Roadmaps
- About
- Login
- Get Started
- Hero section
- Platform introduction
- Feature highlights
- Social proof
- Platform statistics

5.2 Registration

Implement student registration with:

- Full Name
- Email
- Password
- Confirm Password
- Terms & Conditions
- Privacy Policy
- Google Sign-In option

5.3 Login

Implement:

- Student role selection
- Email
- Password
- Remember Me
- Forgot Password
- Google Sign-In
- Registration navigation

5.4 Authentication Backend

Implement:

- User registration API
- Login API
- Password hashing

- Credential validation
- JWT generation
- JWT verification
- Authentication middleware
- Token expiration

5.5 Authorization

Establish role-based authorization architecture for:

- Student
- Mentor
- Admin

5.6 Security

Implement:

- Secure password storage
- JWT signing
- Protected APIs
- Input validation
- Authentication error handling
- Authorization checks

Milestone 1 Deliverable

A user can:

Register → Login → Receive JWT → Access Protected Resources → Logout

6. Milestone 2 – Student Dashboard

Objective

Create the personalized workspace for authenticated students.

6.1 Student Dashboard

Develop:

- Personalized welcome section
- Learner XP
- Active courses
- Completed courses
- Study hours
- XP points
- Learning streak
- Achievements
- Certificates

6.2 Course Overview

Implement:

- My Enrolled Courses
- Course thumbnails
- Course categories
- Course progress
- Completion percentage
- Continue Course
- Certificate indicator

6.3 Learning Progress

Implement:

- Weekly progress
- Learning activity
- Course completion
- Study hours
- Learning streak

6.4 Achievements

Implement:

- Recent achievements
- Achievement icons
- Achievement descriptions
- Achievement status

6.5 Notifications

Implement:

- Student notifications
- Read/unread status
- Course notifications
- Achievement notifications
- Certificate notifications

6.6 Recommendations

Implement:

- Recommended courses
- Category-based recommendations
- Career-related recommendations

6.7 Navigation

Create student navigation for:

- Dashboard
- Catalog
- My Courses
- AI Tutor
- Roadmap
- Skill Gap
- Resume
- Practice
- Assignments
- Quizzes
- Certificates
- Careers
- Sessions
- Analytics
- Notifications
- Profile
- Settings

Milestone 2 Deliverable

An authenticated student receives a personalized dashboard containing their learning and career information.

7. Milestone 3 – Admin & Mentor Dashboards

Objective

Introduce separate operational dashboards for administrators and mentors.

7.1 Admin Dashboard

Develop:

- Welcome section
- Total Users
- Students
- Mentors
- Admins
- Total Courses
- Pending Approvals
- Active Courses
- Complaints

7.2 Admin Management

Implement:

User Management

- View users
- Search users
- View user details
- Update users
- Manage user status

Role Management

- Student
- Mentor
- Admin

Course Approvals

- View pending courses
- Review course
- Approve course
- Reject course

Complaint Desk

- View complaints
- Update complaint status
- Resolve complaints
- Track complaint history

Audit Logs

Record:

- Administrative actions
- Course approvals
- Complaint updates
- User changes
- System events

Admin Notifications

Display:

- Approval requests
 - System alerts
 - Complaints
 - Platform events
-

8. Mentor Dashboard

8.1 Mentor Overview

Develop:

- Total Students
- Courses Created
- Pending Assignments
- Pending Quizzes
- Published Courses
- Draft Courses
- Total Enrollments

8.2 Course Management

Mentors can:

- Create courses
- Edit courses
- Save drafts
- Submit courses for approval
- View course status
- Manage published courses

8.3 Student Activity

Display:

- New enrollments
- Assignment submissions
- Course activity
- Student progress

8.4 Assignments

Implement:

- Assignment creation
- Assignment editing
- Submission viewing
- Grading
- Feedback

8.5 Quizzes

Implement:

- Quiz creation
- Question management
- Quiz submission
- Result management
- Score tracking

8.6 Sessions

Implement:

- Create sessions
- Schedule sessions
- View upcoming sessions
- Edit sessions
- Join session functionality

8.7 Mentor Notifications

Support:

- New enrollment
- Assignment submission
- Quiz activity
- Course approval
- Upcoming session

Milestone 3 Deliverable

The platform supports three distinct role experiences:

Student + Mentor + Admin

with appropriate navigation and permissions.

9. Milestone 4 – Backend & Full System Integration

Objective

Connect all frontend components to real backend services and database systems.

This milestone converts the developed UI into a fully functional application.

10. Backend Development

Develop and integrate backend services for:

- Authentication
 - Users
 - Roles
 - Courses
 - Lessons
 - Enrollments
 - Progress
 - Assignments
 - Quizzes
 - Sessions
 - Achievements
 - Certificates
 - Notifications
 - Complaints
 - Approvals
 - Internships
 - Audit logs
 - Search
 - Dashboard statistics
-

11. Database Integration

Create and connect database models for:

User

- ID
- Name
- Email
- Password Hash
- Role
- Status
- Created Date
- Updated Date

Course

- Course ID
- Title
- Description
- Category
- Mentor
- Status
- Duration
- Created Date

Enrollment

- Student
- Course
- Enrollment Date
- Progress
- Completion Status

Assignment

- Assignment
- Course
- Student
- Submission
- Grade
- Feedback
- Status

Quiz

- Quiz
- Questions
- Attempts
- Score
- Result

Achievement

- Student
- Achievement
- Description
- Date Earned

Certificate

- Student
- Course
- Certificate ID
- Issue Date
- Verification Information

Notification

- Recipient
- Notification Type
- Message
- Read Status
- Timestamp

Audit Log

- User
- Action
- Resource
- Description
- Timestamp

12. API Integration

Connect frontend components with backend APIs.

Authentication APIs

- Register
- Login
- Logout
- Forgot Password
- Google Authentication

Student APIs

- Student Profile
- Dashboard
- Courses
- Enrollments
- Progress
- Achievements
- Certificates
- Notifications
- Recommendations

Mentor APIs

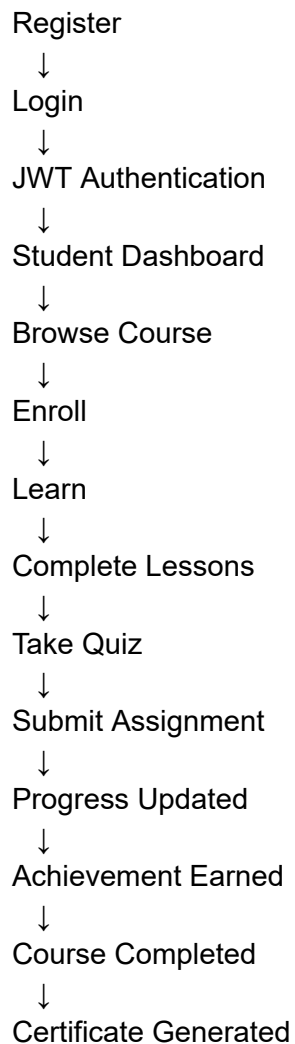
- Mentor Dashboard
- Courses
- Students
- Assignments
- Quizzes
- Grading
- Sessions
- Enrollments

Admin APIs

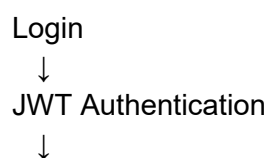
- Admin Dashboard
- Users
- Roles
- Approvals
- Complaints
- Audits
- Internships
- Notifications

13. End-to-End Workflows

Student Workflow



Mentor Workflow



Mentor Dashboard
↓
Create Course
↓
Save Draft
↓
Submit for Approval
↓
Admin Review
↓
Course Approved
↓
Course Published
↓
Students Enroll
↓
Monitor Students
↓
Review Assignments
↓
Grade Students
↓
Manage Quizzes
↓
Schedule Sessions

Admin Workflow

Login
↓
JWT Authentication
↓
Admin Dashboard
↓
Monitor Platform
↓
Manage Users
↓
Manage Roles
↓
Review Courses
↓

Approve / Reject



Monitor Complaints



Review Audit Logs



Monitor Platform Operations

14. Role & Permission Matrix

Feature	Student	Mentor	Admin
Student Dashboard	✓	-	-
Mentor Dashboard	-	✓	-
Admin Dashboard	-	-	✓
View Courses	✓	✓	✓
Create Course	-	✓	✓
Approve Course	-	-	✓
Enroll Course	✓	-	✓
Manage Students	-	✓	✓
Assignments	✓	✓	✓
Grading	-	✓	✓
Quizzes	✓	✓	✓
Sessions	✓	✓	✓
Certificates	✓	✓	✓
User Management	-	-	✓
Role Management	-	-	✓

Complaints	Submit/View	Relevant	Manage
Audit Logs	-	-	✓
Platform Settings	-	Limited	✓

15. Security Plan

Security is implemented across all layers.

Authentication

- JWT authentication
- Secure password hashing
- Token expiration
- Protected routes

Authorization

- Role-based access control
- API permission checks
- Resource ownership validation

Data Security

- Passwords never stored as plain text
- Sensitive credentials stored securely
- Input validation
- Secure database queries
- HTTPS in production

API Security

- Authentication middleware
- Authorization middleware
- Request validation
- Error handling
- Rate limiting where required

16. Testing Strategy

Testing should be performed at multiple levels.

Unit Testing

Test individual:

- Functions
- Services
- Controllers
- Validation functions
- Authentication logic

API Testing

Test:

- Registration
- Login
- JWT validation
- Course APIs
- Enrollment APIs
- Assignment APIs
- Quiz APIs
- Admin APIs
- Mentor APIs

Integration Testing

Verify:

Frontend → API → Backend → Database

Role Testing

Verify:

- Student permissions

- Mentor permissions
- Admin permissions

UI Testing

Verify:

- Forms
 - Buttons
 - Navigation
 - Dashboard
 - Responsive layout
 - Error messages
-

17. Security Testing

Security testing should verify:

- Invalid JWT rejection
- Expired JWT rejection
- Unauthorized API rejection
- Role escalation prevention
- Password security
- Input validation
- Protected resources
- User data isolation

Example:

A Student attempting to access an Admin API must be rejected by the backend even if the student manually sends the request.

18. Performance Testing

The system should be evaluated for:

- API response time
- Dashboard loading time

- Database query performance
- Search performance
- Concurrent users
- Large course lists
- Large notification lists

Optimization may include:

- Database indexing
 - Pagination
 - API response optimization
 - Caching where appropriate
 - Lazy loading
 - Query optimization
-

19. Error & Exception Handling

The platform should provide consistent handling for:

- Invalid requests
- Invalid credentials
- Expired sessions
- Unauthorized access
- Missing records
- Duplicate records
- Database failures
- API failures
- Network failures

Frontend should display understandable messages while backend logs should retain useful technical information for debugging.

20. Deployment Plan

Development Environment

Develop and test features locally.

Testing/Staging Environment

Deploy the integrated system for:

- QA testing
- API testing
- Security testing
- User acceptance testing

Production Environment

Deploy:

- Frontend
- Backend
- Database
- Authentication services
- Required storage/services

Production configuration should use secure environment variables.

Sensitive values should not be committed to source control.

21. Documentation Plan

The project documentation should contain:

Technical Documentation

- System architecture
- Database schema
- API documentation
- Authentication architecture
- Authorization model
- Deployment instructions

User Documentation

- Student guide
- Mentor guide

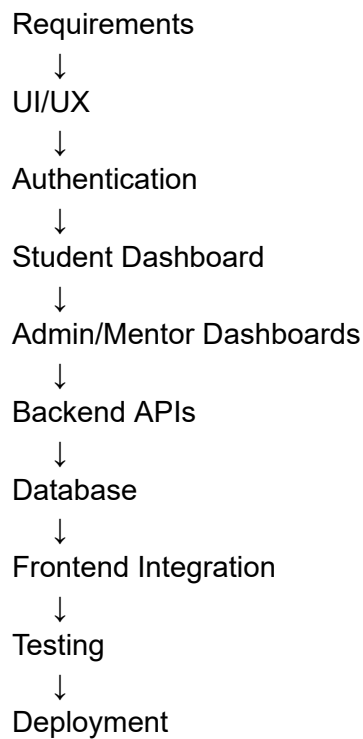
- Admin guide

Project Documentation

- Requirements
 - Milestones
 - Testing results
 - Known limitations
 - Future enhancements
-

23. Project Dependencies

The development sequence follows these dependencies:



Authentication must be completed before protected dashboards can be fully integrated.

Database and API development must be available before dashboard data can become dynamic.

24. Project Deliverables

At project completion, the following deliverables should be available:

Application

- Student portal
- Mentor portal
- Admin portal
- Authentication system
- Dashboard system
- Course system
- Assessment system
- Career guidance system
- Internship system
- Certificate system
- Notification system

Backend

- REST/API services
- Authentication
- Authorization
- Business logic
- Database integration
- Validation
- Error handling

Database

- Complete schema
- Relationships
- Constraints
- Indexes
- Seed/initial data where required

Security

- JWT authentication
- Password hashing
- Role-based authorization

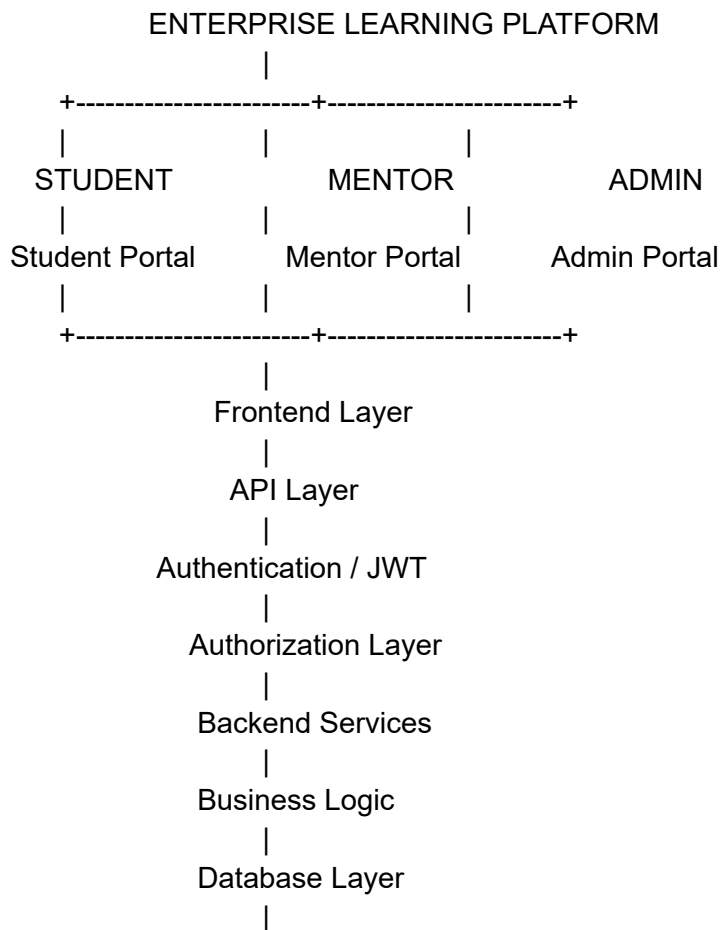
- Protected APIs
- Secure configuration

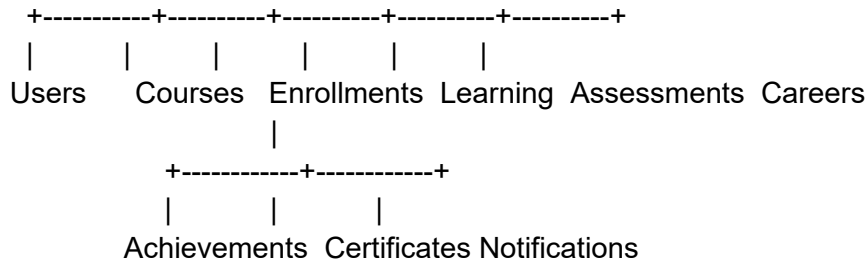
Documentation

- Project report
 - API documentation
 - Database documentation
 - User documentation
 - Deployment documentation
 - Testing report
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25. Final Project Architecture

The final system can be represented as:





26. Milestone Summary

Milestone	Main Objective	Key Deliverables
Milestone 1	Foundation & Authentication	Landing page, Login, Registration, JWT, Authorization
Milestone 2	Student Experience	Student Dashboard, Courses, Progress, Achievements, Certificates
Milestone 3	Platform Operations	Admin Dashboard, Mentor Dashboard, Management Tools
Milestone 4	Full Integration	Backend, APIs, Database, End-to-End Integration
Future	Advanced Platform	AI Tutor, Skill Gap, Roadmaps, Careers, Advanced Analytics

27. Current Project Status

Based on the implemented milestones:

Completed

Milestone 1

- Platform foundation
- Authentication UI
- Registration
- Login

- JWT authentication
- Authorization foundation

Milestone 2

- Student Dashboard
- Student statistics
- Course overview
- Progress
- Achievements
- Certificates
- Notifications
- Recommendations

Milestone 3

- Admin Dashboard
- Mentor Dashboard
- User management
- Course management
- Approvals
- Complaints
- Audit logs
- Assignments
- Quizzes
- Sessions
- Mentor activity

Milestone 4

- Backend integration
- API integration
- Database integration
- Authentication integration
- Authorization integration
- Dashboard integration
- End-to-end data flow

29. Final Project Goal

The final system should provide an integrated ecosystem where:

Students learn and develop skills



Mentors create and deliver learning experiences



Administrators manage and govern the platform



The system tracks learning activity



Skills and gaps are identified



Career paths are recommended



Students receive internships and opportunities



Achievements and certificates validate their progress

Final Project Statement

The Enterprise Learning Platform is being developed as a **secure, role-based, full-stack learning and career guidance ecosystem**.

The first four milestones establish the core technical foundation:

Milestone 1: Authentication & Security

Milestone 2: Student Experience

Milestone 3: Admin & Mentor Operations

Milestone 4: Backend & Full-System Integration

Together, these milestones establish the foundation required to develop the advanced **learning, skill assessment, career guidance, AI, internship, analytics, and certification capabilities** of the complete platform.

